MTH108 2023S1 Exam

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH108

Examination - January Semester 2023

Calculus II

Tuesday, 23 May 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total 100 marks)

Question 1

Use various techniques of integration to evaluate the following integrals. Show your working clearly.

Question 1a

$$\int 2023^{x^2} x \, dx$$

(5 marks)

Question 1b

$$\int_0^{\frac{1}{3}} \sin^{-1}(3x) \; dx$$

(8 marks)

Question 1c

$$\int \frac{3x^2 - x + 2}{(x-2)(x^2-4)} \; dx$$

(7 marks)

Question 2

Let R be the region bounded by the curve $y = x^2 - 4x$ and the line $y = x - 4$.

Question 2a

Find the area of the region R. Show your working clearly.

(6 marks)

Question 2b

Apply the disk/washer method to find the <u>exact</u> volume of the solid of revolution obtained when R is rotated 2π radians about the x-axis.

(7 marks)

Question 2c

Using the method of cylindrical shells, find the <u>exact</u> volume of the solid of revolution obtained when R is rotated 2π radians about the line $x = 4$.

(7 marks)

Question 3a

Solve

$$\frac{d}{dx}\int_{x^e}^{e^x} \ln t \, dt \, .$$

Show your working clearly.

(7 marks)

Question 3b

Define the function $f : \mathbb{R}\backslash\{\tan\frac{1}{2}\} \longrightarrow \mathbb{R}$ by

$$f(x) = \frac{1 + \tan^{-1} x}{1 - 2\tan^{-1} x} \, .$$

Determine the derivative of the function f and simplify your answer as much as possible. Show your working clearly.

(6 marks)

Question 3c

Define the function $g : (-\infty, 3) \longrightarrow \mathbb{R}$ by

$$g(x) = 3\ln(3 - x) \, .$$

Determine an expression for $g^{-1}(x)$ as well as its domain and range.

(7 marks)

Question 4a

Let $g : [0, 27] \longrightarrow \mathbb{R}$ be a function which is Riemann integrable. Given that

$$\int_0^{27} g(x)\, dx = 3\,,$$

evaluate

$$\int_0^3 x^2 g(x^3)\, dx\,.$$

(6 marks)

Question 4b

Show that the following reduction formula holds for $n \neq 1$.

$$\int \frac{1}{\sin^n x}\, dx = -\frac{\cos x}{(n-1)\sin^{n-1} x} + \frac{n-2}{n-1}\int \frac{1}{\sin^{n-2} x}\, dx$$

(7 marks)

Question 4c

Interpret

$$\lim_{n\to\infty} \frac{3}{n}\sum_{k=1}^{n} \left(\frac{3k}{n} - 3\right)^2$$

as a definite integral and evaluate it.

(7 marks)

Question 5

Use various techniques of integration to evaluate the following integrals. Show your working clearly.

Question 5a

$$\int 2023^x \cos x \, dx$$

(10 marks)

Question 5b

$$\int_0^3 x^2 \sqrt{9 - x^2} \, dx$$

(10 marks)

----- END OF PAPER -----

1(a) let $e^k = 2023$

$\Rightarrow k = \ln 2023$

$$\int 2023^{x^2} x \, dx = \int (e^k)^{x^2} x \, dx$$
$$= \int x e^{kx^2} dx$$
$$= \frac{1}{2k} \int (2kx) e^{kx^2} dx$$
$$= \frac{1}{2k} e^{kx^2} + C$$
$$= \frac{1}{2\ln 2023} 2023^{x^2} + C$$ where C is a constant

(b)

let $u = 3x$ $\quad \frac{du}{dx} = 3 \Rightarrow dx = \frac{1}{3} du$

$$\int \sin^{-1} 3x \, dx = \int (\sin^{-1} u) \frac{1}{3} du$$
$$= \frac{1}{3} \int \sin^{-1} u \, du$$
$$= \frac{1}{3} \left[u \sin^{-1} u + \sqrt{1-u^2} \right] + C$$ where C is a constant

when $x=0$, $u=0$. when $x=\frac{1}{3}$, $u=1$

$$\int_0^{\frac{1}{3}} \sin^{-1} 3x \, dx = \frac{1}{3} \left[u \sin^{-1} u + \sqrt{1-u^2} \right]_{u=0}^{u=1}$$
$$= \frac{1}{3} \left[\sin^{-1} 1 - \sqrt{1} \right]$$
$$= \frac{\pi}{6} - \frac{1}{3}$$

(c)

$$\frac{3x^2 - x + 2}{(x-2)(x^2-4)} = \frac{3x^2 - x + 2}{(x-2)(x-2)(x+2)}$$

let $$\frac{3x^2 - x + 2}{(x-2)^2(x+2)} = \frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$\Leftrightarrow 3x^2 - x + 2 = A(x-2)^2 + B(x+2)(x-2) + C(x+2)$$
$$= A(x^2 - 4x + 4) + B(x^2 - 4) + C(x+2)$$
$$= (A+B)x^2 + (C-4A)x + (4A - 4B + 2C)$$

$\Leftrightarrow A + B = 3 \quad C - 4A = -1, \quad 4A - 4B + 2C = 2$

$B = 3 - A$

$C = 4A - 1$

$2 = 4A - 4B + 2C$

$= 4A - 4(3-A) + 2(4A-1)$

$= 16A - 14$

$16A = 16$

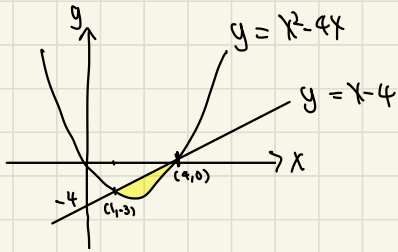
$A = 1$

$B = 3 - A = 2$

$C = 4A - 1 = 3$

$$\therefore \int \frac{3x^2 - x + 2}{(x-2)(x^2-4)} dx$$
$$= \int \frac{1}{x+2} + \frac{2}{x-2} + \frac{3}{(x-2)^2} dx$$
$$= \ln(x+2) + 2\ln(x-2) - \frac{3}{x-2} + K,$$
where K is a constant

2(a)



let $x^2 - 4x = x - 4$

$x^2 - 5x + 4 = 0$

$(x-4)(x-1) = 0$

$x = 1$ or 4

$y = -3$ or 0

area of $R = \int_1^4 (x^2 - 4x) - (x - 4)\, dx$

$= \int_1^4 x^2 - 5x + 4 \, dx$

$= \left[\frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x \right]_1^4$

$= 3$

(b) required volume

$= \int_1^4 \pi r_2^2(x) - \pi r_1^2(x)\, dx$ where $r_1(x) = 0 - (x-4) = 4 - x$

$r_2(x) = 0 - (x^2 - 4x) = 4x - x^2$

$= \int_1^4 \pi (4x - x^2)^2 - \pi (4 - x)^2 \, dx$

$= \pi \int_1^4 x^4 - 8x^3 + 15x^2 + 8x - 16 \, dx$

$= \pi \left[\frac{1}{5}x^5 - 2x^4 + 5x^3 + 4x^2 - 16x \right]_1^4$

$= \frac{106}{5}\pi$

(c) required volume

$= \int_1^4 2\pi r(x) h(x)\, dx$ where $r(x) = 4 - x$, $h(x) = (x-4) - (x^2 - 4x) = 5x - x^2 - 4$

$= 2\pi \int_1^4 (4 - x)(5x - x^2 - 4) \, dx$

$= 2\pi \int_1^4 x^3 - 9x^2 + 24x - 16 \, dx$

$= 2\pi \left[\frac{1}{4}x^4 - 3x^3 + 12x^2 - 16x \right]_1^4$

$= \frac{27}{2}\pi$

3(a) $\int \ln t \, dt = t\ln t - \int t\left(\frac{1}{t}\right) dt$

$= t\ln t - \int 1 \, dt$

$= t\ln t - t$

$= t(\ln t - 1)$

$\int_{x^e}^{e^x} \ln t \, dt = \left[t(\ln t - 1)\right]_{x^e}^{e^x}$

$= e^x(\ln e^x - 1) - x^e(\ln x^e - 1)$

$= e^x(x-1) - x^e(\ln x^e - 1)$

$= e^x(x-1) - x^e(e\ln x - 1)$

$\frac{d}{dx}\int_{x^e}^{e^x} \ln t \, dt = \frac{d}{dx}\left[e^x(x-1) - x^e(e\ln x - 1)\right]$

$= \left[e^x(x-1) + e^x\right] - \left[ex^{e-1}(e\ln x - 1) + x^e\left(\frac{e}{x}\right)\right]$

$= xe^x - \left[ex^{e-1}(\ln x^e - 1) + ex^{e-1}\right]$

$= xe^x - ex^{e-1}(\ln x^e)$

(b) let $g(x) = \frac{1+x}{1-2x}$, $y = \tan^{-1} x$

$\frac{dy}{dx} = \frac{1}{1+x^2}$

$\frac{dg}{dx} = \frac{d}{dx}(1+x)(1-2x)^{-1}$

$= (1+2x)^{-1} - (-2)(1+x)(1-2x)^{-2}$

$= \frac{1}{1-2x} + \frac{2(1+x)}{(1-2x)^2}$

$f(x) = g(y)$

$f'(x) = \frac{dg}{dy} \cdot \frac{dy}{dx}$

$= \left[\frac{1}{1-2\tan^{-1}x} + \frac{2(1+\tan^{-1}x)}{(1-2\tan^{-1}x)^2}\right] \times \frac{1}{1+x^2}$

3(c) domain of $g(x) = (-\infty, 3)$
range of $g(x) = (-\infty, \infty)$

let $y = 3\ln(3-x)$

$$\frac{y}{3} = \ln(3-x)$$

$$e^{\frac{y}{3}} = 3-x$$

$$x = 3 - e^{\frac{y}{3}}$$

$$g^{-1}(x) = 3 - e^{\frac{x}{3}}$$

domain of $g^{-1}(x)$ = range of $g(x)$
$= (-\infty, \infty)$

range of $g^{-1}(x)$ = domain of $g(x)$
$= (-\infty, 3)$

4(a) let $G(u) = \int g(u)\,du$
let $u = x^3$

$$\frac{d}{dx}G(u) = \frac{dG}{du}\cdot\frac{du}{dx}$$

$$= g(u)\cdot(3x^2)$$

$$= 3x^2\, g(x^3)$$

$$\int_0^3 x^2 g(x^3)\,dx = \frac{1}{3}\int_0^3 3x^2 g(x^3)\,dx$$

$$= \frac{1}{3}\left[G(u)\right]_0^{3^3}$$

$$= \frac{1}{3}\left[G(u)\right]_0^{27}$$

$$= \frac{1}{3}\int_0^{27} g(u)\,du$$

$$= \frac{1}{3}(3)$$

$$= 1$$

4(b) $\int \frac{1}{\sin^n x}\,dx = \int \operatorname{cosec}^n x\,dx$

$$= \int \operatorname{cosec}^2 x \operatorname{cosec}^{n-2} x\,dx$$

$$= (\operatorname{cosec}^{n-2} x)(-\cot x) - \int (n-2)(\operatorname{cosec}^{n-3} x)(-\operatorname{cosec} x \cot x)(-\cot x)\,dx$$

$$= -\frac{1}{\sin^{n-2} x}\cdot\frac{\cos x}{\sin x} - (n-2)\int \frac{1}{\sin^{n-3} x}\cdot\frac{1}{\sin x}\cdot\frac{\cos x}{\sin x}\cdot\frac{\cos x}{\sin x}\,dx$$

$$= -\frac{\cos x}{\sin^{n-1} x} - (n-2)\int \frac{\cos^2 x}{\sin^n x}\,dx$$

$$= -\frac{\cos x}{\sin^{n-1} x} - (n-2)\int \frac{1-\sin^2 x}{\sin^n x}\,dx$$

$$= -\frac{\cos x}{\sin^{n-1} x} - (n-2)\int \frac{1}{\sin^n x} - \frac{1}{\sin^{n-2} x}\,dx$$

$$= -\frac{\cos x}{\sin^{n-1} x} + (n-2)\int \frac{1}{\sin^{n-2} x}\,dx - (n-2)\int \frac{1}{\sin^n x}\,dx$$

$$(n-1)\int \frac{1}{\sin^n x}\,dx = -\frac{\cos x}{\sin^{n-1} x} + (n-2)\int \frac{1}{\sin^{n-2} x}\,dx$$

$$\therefore \int \frac{1}{\sin^n x}\,dx = -\frac{\cos x}{(n-1)\sin^{n-1} x} + \frac{n-2}{n-1}\int \frac{1}{\sin^{n-2} x}\,dx$$

4(c) $\lim_{n\to\infty} \frac{3}{n}\sum_{k=1}^{n}\left(\frac{3k}{n}-3\right)^2 = \lim_{n\to\infty}\sum_{k=1}^{n}(x_k-3)^2\,\Delta x$ where $x_k = \frac{3k}{n}$, $\Delta x = \frac{3}{n}$

$$= \int_0^3 (x-3)^2\,dx$$

$$= \left[\frac{1}{3}(x-3)^3\right]_0^3$$

$$= 9$$

5(a) let $2023 = e^k$ where $k = \ln 2023$

$$\int 2023^x \cos x \, dx = \int \underbrace{e^{kx}}_{dv} \underbrace{\cos x}_{u} \, dx$$

use LIATE rule for integration by parts

$$= (\cos x)\left(\tfrac{1}{k}e^{kx}\right) - \int (-\sin x)\left(\tfrac{1}{k}e^{kx}\right) dx$$

$$= \tfrac{1}{k}e^{kx}\cos x + \tfrac{1}{k^2}\int \underbrace{(\sin x)}_{u}\underbrace{(ke^{kx})}_{dv}\, dx$$

(integration by parts)

$$= \tfrac{1}{k}e^{kx}\cos x + \tfrac{1}{k^2}\left[e^{kx}\sin x - \int e^{kx}\cos x \, dx\right]$$

$$= \tfrac{1}{k}e^{kx}\cos x + \tfrac{1}{k^2}e^{kx}\sin x - \tfrac{1}{k^2}\int e^{kx}\cos x \, dx + C_1$$

where C_1 is a constant

$$\left(1+\tfrac{1}{k^2}\right)\int e^{kx}\cos x \, dx = \tfrac{1}{k}e^{kx}\cos x + \tfrac{1}{k^2}e^{kx}\sin x + C_1$$

$$(k^2+1)\int e^{kx}\cos x \, dx = ke^{kx}\cos x + e^{kx}\sin x + k^2 C_1$$

$$\int e^{kx}\cos x \, dx = \frac{e^{kx}}{k^2+1}(k\cos x + \sin x) + \frac{k^2 C_1}{k^2+1}$$

$$\therefore \int 2023^x \cos x \, dx = \frac{2023^x}{(\ln 2023)^2+1}\left((\ln 2023)\cos x + \sin x\right) + C_2$$

where C_2 is a constant

$$C_2 = \frac{k^2 C_1}{k^2+1}$$

5(b) let $x = 3\cos(u)$, $u = \cos^{-1}\frac{x}{3}$

$\frac{dx}{du} = -3\sin(u)$, $dx = -3\sin(u)\,du$

$$\int x^2\sqrt{9-x^2}\,dx = \int (3\cos u)^2\sqrt{9-9\cos^2 u}\,(-3\sin u)\,du$$

$$= \int -9\sin u\cos^2 u\sqrt{9(1-\cos^2 u)}\,du$$

$$= \int -9\sin u\cos^2 u\sqrt{9\sin^2 u}\,du$$

$$= \int -9\sin u\cos^2 u\,(3\sin u)\,du$$

$$= -27\int \sin^2 u\cos^2 u\,du$$

$$= -\frac{27}{4}\int 4\sin^2 u\cos^2 u\,du$$

$$= -\frac{27}{4}\int (2\sin u\cos u)^2\,du$$

$\sin 2u = 2\sin u\cos u$

$\sin^2 2u = 4\sin^2 u\cos^2 u$

$$= -\frac{27}{4}\int \sin^2 2u\,du$$

$$= -\frac{27}{8}\int 2\sin^2 2u\,du$$

$$= -\frac{27}{8}\int 1-\cos 4u\,du$$

$\cos 2A = 1-2\sin^2 A$

where $A = 2u$

$$= -\frac{27}{8}\left[u-\frac{1}{4}\sin 4u\right]+C$$ where C is a constant

$$\int_0^3 x^2\sqrt{9-x^2}\,dx = -\frac{27}{8}\left[u-\frac{1}{4}\sin 4u\right]_{u_1}^{u_2}$$ where $u_1 = \cos^{-1}\frac{0}{3} = \cos^{-1}0 = \frac{\pi}{2}$

$u_2 = \cos^{-1}\frac{3}{3} = \cos^{-1}1 = 0$

$$= \frac{27}{8}\left[u-\frac{1}{4}\sin 4u\right]_{u_2}^{u_1}$$

$$= \frac{27}{8}\left[u-\frac{1}{4}\sin 4u\right]_0^{\frac{\pi}{2}}$$

$$= \frac{27}{16}\pi$$